



FIRNANDA SURYA HERDIANSYAH

0895638019956 | firnandas081@gmail.com | LinkedIn : <https://www.linkedin.com/in/firnanda-surya-herdiansyah-2b1ab5320/> | IG : @firnandasurya18 | Madiun, 10 Januari 2002

Desa Sobrah RT 02 / RW 01 Kec. Wungu Kab. Madiun

I am a Bachelor of Piping Engineering at Surabaya State Shipbuilding Polytechnic, with a strong interest in the EPC (Engineering, Procurement, and Construction) field. Familiarity with industry standards (ASME, API, ANSI, etc.) I am highly familiar with Piping and Pipeline engineering software such as AutoCAD, Applied Flow Technology, and Piping Stress Analysis. I am able to maintain a professional attitude both when working individually and as part of a team, and I have a strong desire to learn new things in engineering technology. Additionally, I have experience in construction and power plants, and Oil and Gas Project.

Work Experiences

PT. MIPCON PRIMA INDUSTRI – PERTAMINA ZONA 1 JAMBI FIELD **Jakarta**

Mechanical Piping Engineer (Project Management Team) | May 2025 – Now

- Develop piping system designs based on project specifications, P&ID, and engineering standards.
- Prepare layout drawings, isometrics, and support designs.
- Select appropriate materials and components for piping systems according to ASME, ASTM, or project standards.
- Create and review technical documentation such as material take-offs (MTO), bill of materials (BOM), and datasheets.
- Prepare piping specifications, standard drawings, and engineering deliverables.
- Conduct design review and provide technical clarification to vendors and construction teams.
- Coordinate with process, mechanical, structural, and instrumentation teams to ensure piping integration.
- Liaise with procurement for piping materials and equipment.
- Support construction and commissioning teams by resolving technical queries (TQ).
- Review and implement inspection and test plans (ITPs) for piping works.
- Participate in FAT, SAT, pressure testing, and mechanical completion.
- Perform site visits for piping layout verification, field measurements, and troubleshooting.
- Oversee installation, alignment, hydrotest, and inspection of piping systems.
- Track progress of piping deliverables in accordance with project schedule.
- Assist in cost estimation, scheduling, and resource planning for piping scopes.
- Prepare reports for project managers and stakeholders.

PT. Meitech Eka Bintan (AOI SISI NUBI Package A & B) – **Bintan**

Piping Pre-Comm Engineer | January 2025 – May 2025

- Prepare test packs and review P&ID, isometric drawings, and test limits.
- Ensure compliance with project specifications, standards, and procedures.
- Coordinate with construction, commissioning, QA/QC, and operations teams.
- Identify, document, and coordinate the closure of punch list items.
- Conduct hydrostatic testing, pneumatic testing, flushing, and drying of piping systems.
- Oversee flange management program to ensure leak-free start-up.
- Ensure correct flange alignment, gasket selection, and bolt tightening procedures.
- Prepare pre-commissioning reports, test records, and as-built drawings.
- Ensure seamless handover from construction to commissioning.

PT. Brantas Abipraya (Bener Dam Construction Project) - **Purworejo**

BIM Coordinator | August 2024 – January 2025

- BIM Model Coordination
- Managing data and information flow among various stakeholders (owner, contractor, consultant).
- Conducting regular BIM model reviews to ensure accuracy and quality.
- Preparing progress reports for the BIM model and relevant documentation.
- Generating clash detection reports and proposing solutions for resolution.
- Supervising the use of the Common Data Environment (CDE) for secure document sharing.
- Supporting the implementation of BIM during the construction phase to ensure work aligns with the model.

**PT. Brantas Abipraya (Bener Dam Construction Project) -
Purworejo**

Pipeline Engineer | Agustus 2024 – January 2025

- Designing hydromechanical components such as sluice gates, valves, pumps, and turbines for dam projects, hydropower plants, or irrigation systems.
- Performing technical calculations (pressure analysis, force, and flow rate) to ensure the performance and reliability of components.
- Coordinating hydromechanical work with other disciplines such as civil, structural, and electrical engineering.
- Developing hydromechanical work plans, including schedules, budgets, and resource allocation.
- Selecting appropriate materials for hydromechanical components based on operating conditions, such as resistance to corrosion and high pressure.
- Preparing technical reports, including test results, performance analysis, and installation documentation.
- Ensuring all installations comply with quality standards, safety regulations, and design specifications.

**PT. Brantas Abipraya (Bener Dam Construction Project) -
Purworejo**

Hydromechanical Engineer (Internship Trainee) | February 2023 – December 2023

- Make and review pipeline drawings (3D drawings, asbuilt drawings, general arrangements, etc.)
- Piping routing design
- Piping data management and analysis
- Make a list of Material Take-off
- Prepare and create daily progress report
- reviewing NDT documents on bottom pipe connection installation
- Calculate material strength using the standards used and the Finite Element Method

Education Level

Politeknik Perkapalan Negeri Surabaya - Surabaya

- Major : Marine Engineering (Piping Engineering) | August 2020 - August 2024 , GPA : 3,39/4

SMA Negeri 1 Mejayan

- Major : MIPA | 2017 - 2020

SMP Negeri 2 Nglames

- 2014 - 2017

Organisation, Skill & Language

Himpunan Mahasiswa Teknik Perpipaan - Surabaya

Head of Research and Science Department | June 2022 - December 2022

Tim Kapal Emas Politeknik Perkapalan Negeri Surabaya - Surabaya

Research Coordinator and scientific work | June 2022 - December 2022

Himpunan Mahasiswa Teknik Perpipaan - Surabaya

Young Staff of Research and Science Department | October 2021 - June 2022

Soft Skill

- Team Communication and Collaboration
- Problem Solving and Critical Thinking
- Time and Priority Management
- Attention to Detail and Decision Making

Hard Skill

- Ansys
- Naviswork
- Caesar II
- Autocad 3D Plant
- AutoCad
- Applied Flow Technology
- Microsoft Office
- Mapping Software

Language

- Indonesian (Active)
- English (passive)

Certificate of Competency

- **Certified** : Profesional Piping Designers Level 1 Society of Piping Engineers and Designers in 2024.
- **Certified** : Inspektur Pipa Penyalur Minyak Bumi dan Gas (BNSP) in 2024.
- **Certified** : Commisioning Process for Oil and Gas Facilities in 2026.
- **Certified** : Pressure Vessel Design & Calculation As Per ASME Section VIII Using PVElite in 2026.



PPNS
POLITEKNIK
PERKAPALAN
NEGERI SURABAYA

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
POLITEKNIK PERKAPALAN NEGERI SURABAYA

PIN : 005014363022024100034

Terakreditasi A oleh BAN PT, SK No. 1163/SK/BAN-PT/Ak.Ppj/PT/XII/2023

memberikan

IJAZAH

kepada :

Firnanda Surya Herdiansyah

Nomor Registrasi Mahasiswa : 0820040040

Nomor Induk Kependudukan : 3519071001020002

Lahir di Madiun pada tanggal 10 Januari 2002

pada tanggal 29 Agustus 2024 telah menyelesaikan **Pendidikan Sarjana Terapan**

PROGRAM STUDI TEKNIK PERPIPAAN

Terakreditasi UNGGUL oleh LAM Teknik, SK No. 0099/SK/LAM Teknik/VST/VIII/2022

berhak menyandang gelar

SARJANA TERAPAN TEKNIK (S.Tr.T.)

Diterbitkan di Surabaya pada tanggal 17 September 2024

Ketua Jurusan
Teknik Permesinan Kapal

Dr. Priyo Agus Setiawan, ST., MT.
NIP 197708192005011001



Direktur

Rachmad Tri Soelistijono, ST., M.T.
NIP 196811091995121001

TRANSKRIP AKADEMIK

PROGRAM STUDI D4 - TEKNIK PERPIPAAN

Ijin Pendirian: 1639/D/T/2008 - Akreditasi : Unggul (No. 0099/SK/LAM Teknik/VST/VIII/2022)

NOMOR : T/255/PL19/DT.00.00/2024
NAMA : Firnanda Surya Herdiansyah
NRP : 0820040040

TEMPAT/TGL LAHIR : Madiun, 10 Januari 2002
TANGGAL LULUS : 29 Agustus 2024
PIN : 005014363022024100034

NO	KODE	MATA KULIAH	SKS	NILAI	NO	KODE	MATA KULIAH	SKS	NILAI
1	BM41080	Kewirausahaan	2.0	B-	34	PE42200	Sistem Pengolahan Limbah	2.0	B
2	KK40010	Keselamatan dan Kesehatan Kerja	1.0	A-	35	PE42210	Ekonomi Teknik	2.0	A-
3	KK40020	Pengantar Teknologi Kelautan	1.0	A-	36	PE42220	Metode Numerik	2.0	B+
4	KK40030	Etika Profesi	2.0	C+	37	PE42230	Teknologi Proses	1.0	B+
5	KK40040	Metodologi Penelitian	1.0	A-	38	PE42240	Manajemen Proyek	2.0	B+
6	KU41010	Agama	2.0	B+	39	PE42250	Keandalan, Perawatan dan Manajemen Risiko	2.0	A
7	KU41020	Pancasila	2.0	A-	40	PE42260	Sistem Informasi Proyek Perpipaan	1.0	A
8	KU41030	Kewarganegaraan	2.0	A-	41	PE42270	Getaran Mekanis	3.0	A-
9	KU41040	Bahasa Indonesia	2.0	A	42	PE42280	Metode Elemen Hingga	2.0	B+
10	KU41050	Bahasa Inggris I	2.0	B-	43	PE43010	Kode dan Standar Sistem Perpipaan	2.0	B+
11	KU41060	Bahasa Inggris II	2.0	B	44	PE43020	Dasar Perpipaan Sekitar Equipment	1.0	B+
12	KU41070	Bahasa Inggris III	1.0	B	45	PE43030	Tugas Pemodelan Jaringan Instalasi Perpipaan	1.0	B+
13	KU41090	Matematika Teknik	2.0	B+	46	PE43040	Dasar Dasar Plotplan	2.0	B+
14	PE41080	Matematika Dasar	2.0	A-	47	PE43050	Fabrikasi Perpipaan	4.0	B+
15	PE41100	Fisika Terapan	3.0	A-	48	PE43060	Piping Manajemen	2.0	A
16	PE42010	Pengantar Teknologi Perpipaan	2.0	A-	49	PE43070	Pipeline Engineering	2.0	A-
17	PE42020	Teori Bangunan Kapal	3.0	B	50	PE43080	DFKI Tangki dan Pipa	2.0	B
18	PE42030	Teknologi Mekanik	3.0	A-	51	PE43090	Perancangan Bejana Tekan	2.0	B+
19	PE42040	Gambar Teknik	3.0	A-	52	PE43100	Metode Konstruksi	2.0	A-
20	PE42050	Mekanika Fluida	3.0	A	53	PE43110	Instalasi Perpipaan Pembangkit Daya Uap	2.0	B
21	PE42060	Termodinamika dan Perpindahan Panas	3.0	B-	54	PE43120	Sistem Plumbing	4.0	B+
22	PE42070	Mekanika Teknik	2.0	C+	55	PE43130	Analisis Tegangan Perpipaan	2.0	A+
23	PE42080	Gambar Piping General Arrangement dan Isometrik	3.0	B-	56	PE43140	Tugas Perencanaan dan Pengendalian Proyek Perpipaan	2.0	A+
24	PE42090	Ilmu Pengetahuan Bahan	3.0	B	57	PE43150	Inspeksi Perpipaan	3.0	A+
25	PE42100	Praktek Perancangan Dibantu Komputer	1.0	A-	58	PE43160	Tugas Gambar Perpipaan Tingkat Lanjut	3.0	A+
26	PE42110	Statistik	1.0	C+	59	PE43170	Sistem Pemadam Kebakaran	3.0	A+
27	PE42120	Korosi	1.0	A-	60	PE43180	Praktek Kerja Analisa Tegangan Perpipaan	4.0	A+
28	PE42130	Material Perpipaan	2.0	A+	61	PE43190	Praktek Kerja Menggambar Perpipaan Tingkat Lanjut	4.0	B+
29	PE42140	Pneumatik dan Hidrolik	1.0	A	62	PE43200	Praktek Kerja Inspeksi Perpipaan	4.0	B+
30	PE42150	Sistem dalam Kapal	2.0	B	63	PE43210	Praktek Kerja Tugas Perencanaan dan Pengendalian Proyek Perpipaan	4.0	A+
31	PE42160	Pompa dan Kompresor	3.0	A-	64	PE43220	Kapita Selekta	2.0	B
32	PE42170	Teknologi Pengelasan	3.0	A-	65	PE43230	Tugas Rancang Sistem Perpipaan	3.0	B+
33	PE42180	Sistem Pendingin dan Saluran Udara (HVAC)	3.0	B+	66	PE43240	Tugas Akhir	5.0	A

Judul Tugas Akhir

Studi Numerik Pengaruh Closing-Time Valve terhadap Aliran Fluida di Dalam Penstock Pembangkit Listrik Tenaga Air (PLTA) Proyek Pembangunan Bendungan Bener

IPK = $\sum(SKS \cdot NILAI) / \sum SKS$: $511.75 / 151 = 3.39$
Predikat : Sangat Memuaskan
Lama Studi : 8 Semester
Total SKS : 151

Jurusan Teknik Permesinan Kapal
Ketua,



Ekuivalensi Nilai:

A+ : 4 B : 3 D : 1
A : 3.75 B- : 2.75 E : 0
A- : 3.5 C+ : 2.5
B+ : 3.25 C : 2

Surabaya, 17 September 2024
Politeknik Perkapalan Negeri Surabaya
Direktur

Rachman Tri Soelistijono, ST., MT.
NIP 196811081995121001

Dr. Priyo Agus Setiawan, S.T., M.T.
NIP 197708192005011001

13160884



BADAN NASIONAL
SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL
CERTIFICATION AUTHORITY



SERTIFIKAT KOMPETENSI CERTIFICATE OF COMPETENCE

No. 21460 2146 0000184 2024

Dengan ini menyatakan bahwa,
This is to certify that,

Firnanda Surya Herdiansyah

No. Reg. M. 197.16 0022 2024

Telah kompeten pada bidang:
Is competent in the area of:

Jasa Profesional Perpipaan
Piping Professional Services

Dengan Kualifikasi/Kompetensi:
With Qualification/Competency:

Inspektur Pipa Penyalur Minyak Dan Gas Bumi
Channeling Pipe Inspector

Sertifikat ini berlaku untuk: 3 (tiga) tahun
This certificate is valid for: 3 (three) years

Surabaya, 5 November 2024

Atas nama (On Behalf of)
Badan Nasional Sertifikasi Profesi
Indonesia Professional Certification Authority
Lembaga Sertifikasi Profesi PPNS
PPNS Professional Certification Body



Dr. Ir. Wiwik Dwi Pratiwi, M.T.

Ketua
Chairman



Daftar Unit Kompetensi

List of Unit(s) of Competency

KODE UNIT KOMPETENSI <i>CODE OF COMPETENCY UNIT</i>	JUDUL UNIT KOMPETENSI <i>TITLE OF COMPETENCY UNIT</i>
M.712034.001.01	Menerapkan Peraturan dan Perundangan Keselamatan, Kesehatan Kerja dan Lindungan Lingkungan <i>Implementing the Regulation and Law of Occupational Safety, Health and Environmental Protection</i>
M.712034.002.01	Menerapkan Keselamatan Kerja Peralatan di Tempat Kerja <i>Implementing the Occupational Safety at the Workplace</i>
M.712034.003.01	Melakukan Identifikasi Dokumen Perencanaan dan/atau Riwayat Data Pipa Penyalur <i>Performing the Planning Document Identification and/or Data history of Channeling Pipes</i>
M.712034.004.01	Melakukan Identifikasi Pipa Penyalur <i>Performing the Channeling Pipes Identification</i>
M.712034.005.01	Melakukan Inspeksi Fisik Sistem Pipa Penyalur <i>Performing the Physical Inspection of Channeling Pipes System</i>
M.712034.006.01	Melakukan Inspeksi Fisik Komponen Pipa Penyalur <i>Performing the Physical Inspection of Channeling Pipes Component</i>
M.712034.007.01	Melakukan Inspeksi Sistem Pipa Penyalur di Atas Permukaan <i>Performing the Inspection of Above the Surface Channeling Pipes System</i>
M.712034.008.01	Melakukan Inspeksi Sistem Pipa Penyalur di Bawah Tanah <i>Performing the Inspection of Underground Channeling Pipes System</i>
M.712034.009.01	Melakukan Inspeksi Sistem Pipa Penyalur di Bawah Air <i>Performing the Inspection of Under Water Channeling Pipes System</i>
M.712034.010.01	Melakukan Evaluasi Hasil Inspeksi Pipa Penyalur <i>Performing the Evaluation of Channeling Pipes Inspection Results</i>
M.712034.011.01	Membuat Laporan dan Rekomendasi Hasil Inspeksi Pipa Penyalur <i>Creating the Report and Recommendation of Channeling Pipes Inspection Results</i>



Firnanda Surya Herdiansyah

Tanda tangan pemilik
Signature of holder

Surabaya, 5 November 2024
Atas nama (*On Behalf Of*)
Badan Nasional Sertifikasi Profesi
Indonesia Professional Certification Authority
Lembaga Sertifikasi Profesi PPNS
PPNS Professional Certification Body

Mahasin Maulana Ahmad, S.T., M.T.

Manager Sertifikasi
Certification Manager



International

SOCIETY OF PIPING ENGINEERS AND DESIGNERS

Confers the certification of

Professional Piping Designer Level I

To

Surya Herdiansyah Firmanda

Paul Rogers

President

[Signature]

Executive Director

June 02, 2024

PPD Member Since

June 02, 2027

Expiry Date

PPD 2414020

Membership Number



SOCIETY OF PIPING ENGINEERS & DESIGNERS

President: Paul Bowers – PPD Level IV

Executive Director: Catherine VD Walt

Surya Herdiansyah Firnanda – PPD Level I – Grades –

June 02, 2024

PPD Membership Number 2414020

Grade item	Calculated weight	Grade	Range	Percentage	Contribution to course total
<u>Final Exam - PPD Level 1 Part 1</u>	20.00 %	98.40	0-100	98.40 %	19.68 %
<u>Final Exam - PPD Level 1 Part 2</u>	20.00 %	79.37	0-100	79.37 %	15.87 %
<u>Final Exam - PPD Level 1 Part 3</u>	20.00 %	98.63	0-100	98.63 %	19.73 %
<u>Final Exam - Level 1 Part 4</u>	20.00 %	90.95	0-100	90.95 %	18.19 %
<u>Final Exam - Level 1 Part 5</u>	20.00 %	97.30	0-100	97.30 %	19.46 %

Course total
Simple weighted
mean of grades.

92.93 0-100 92.93 %



SOCIETY OF PIPING ENGINEERS & DESIGNERS

President: – Paul Bowers – PPD Level IV

Executive Director: Catherine VD Walt

Transcript for Level I Piping Design and Engineering

Student: **Surya Herdiansyah Firnanda**

Date: **2024-06-02**

PPD Membership Number: 2414020

Grade item Calculated Weight Grade Range Percentage Feedback Contribution to course total

PPD Level I Exam Online

PPD Level 1 Part 1 **98.40 %**

- Introduction to Process Plant Design - Stakeholders, Roles and Results, Project Life Cycle
- Design Phases and Information - Design Phases, Information Flow, Process Documents, Piping Documents

PPD Level 1 Part 2 **79.37 %**

- Pipe Manufacturing & Fabrication - Manufacturing Processes, Pipe Sizes,
- Marking and Coating, Fittings and Components
- Pipe Fabrication - Shop Welding, Branch Connections, Heat Treat and Shipping
- Pipe Assembly - Cranes and Lifts, Scaffolding, Pipe Field Assembly, Pipe Rigging and Lifts, Hydrotest Insulation Labeling
- Typical Fabrication ISOs
- Pipe and Weld Materials - Pipe Materials and Properties, Welding and Welder qualifications, Weld Defects and Inspection

PPD Level 1 Part 3 **98.63 %**

- Valve Types & Applications - Valve Parts, Isolation Valves, Gate Valves, Butterfly Valves, Isolation Applications, Throttling Valves, Valve Actuators
- Self-Actuating Valves - Check Valves, Regulator and Relief Valves, Steam Traps
- Valve Selection and Use - Valve Symbols and Loops, Valve Selection, Station Layout
- Instrumentation - Control Loop Elements, Pressure Measurement, Flow Measurement, Temperature Measurement, Level Measurement, Installation Concepts



SOCIETY OF PIPING ENGINEERS & DESIGNERS

President: – Paul Bowers – PPD Level IV

Executive Director: Catherine VD Walt

PPD Level 1 Part 4

90.95 %

- Pump Selection - Centrifugal Pump Configurations, Centrifugal Pump Cavitation and NPSH, Centrifugal Pump System Design, Positive Displacement Pumps, Motive Power for Pumps
- Pump Layout - Configuration and Elevation, Pump Piping Layout, Temporary Strainers, Pump Access, Pump Loading and Redundancy, Pump Installation
- Pipe Routing - Piping Layout Data, Pipe Routing Considerations, Piping Component Use
- Pipe Racks - Pipe Rack Terminology, Rack and Bent Layout, Rack Pipe Layout, Rack Pipe Expansion, Construction and Details, Other Uses
- Pipe Layout Specification

PPD Level 1 Part 5

97.30 %

- Pressure Vessels - Terminology, Head Shapes and Attachments, Vessel Layouts and Nozzles, ASME Section VIII Concepts, Vessel Fabrication and Erection
- Pipe Supports and Hangers - Span Types and Terminology, Spans and Characteristics, Spring Hanger Placement, Spring Design
- Pipe Stress and B31-3 - Component Class, Pipe Stress Concepts, Weight Solutions for Rack Piping, Thermal Expansion, Simplified Pump Nozzle Load Check, Simplified Vessel Pipe Stress Check, Simplified Expansion Loop Design, ASME B31-3 Concepts
- CAD - Data Structures, Rendering from Data, Database Issues, Database integrity Capture and Reuse

Course total Simple weighted mean of grades.

92.93 %

Certificate ID : CPOG-I-26-EA-0150

ENGINEERING ACADEMY

Powered by Anak Teknik Indonesia

CERTIFICATE OF COMPLETION

This certificate is proudly presented to

Firnanda Surya Herdiansyah

Has successfully completed training of
Commissioning Process For Oil And Gas Facilities
 (Pre-Commissioning, Commissioning, And Start-Up)
Delivered by Amer Hanafiah
(Process/Production SME at IHRDC)

2 days training on 09,10 January 2026
Total duration: 5 hours

Date Issued
 14 January 2026



Arkastra Bhrevijana
 CEO Anak Teknik Indonesia

Amer Hanafiah
 Commissioning Controller



List of learning modules training of **Commissioning Process for Oil and Gas Facilities** (Pre-commissioning, commissioning, and start-up)

NIK INDONESIA • ANAK TEKNIK INDONESIA • ANAK TEKNIK INDONESIA • ANAK TEKNIK INDONESIA • ANAK TEKNIK INDONESIA • ANAK TEKNIK INDONESIA • ANAK TEKNIK INDONESIA

Outline For The Training Program:

1. Introduction To Pre-Commissioning, Commissioning, And Start-Up
2. Mechanical Completion And Pre-Commissioning
3. Commissioning Planning And Preparation
4. Commissioning Execution
5. Start-Up Planning And Preparation
6. Start-Up Execution
7. Post-Commissioning Activities

The Training Program Will Cover The Following Topics:

1. The Importance Of Pre-Commissioning, Commissioning, And Start-Up
2. The Differences Between Pre-Commissioning, Commissioning, And Start-Up
3. The Objectives Of Each Stage
4. The Activities Involved In Each Stage
5. The Challenges And Risks Associated With Eachstage
6. The Best Practices For Each Stage
7. The Roles And Responsibilities Of Thecommissioning Team
8. The Safety Considerations During Each Stage



Learning Outcomes of the Training on Pressure Vessel Design & Calculation as per ASME Section VIII Using PVElite

- 1. ASME Section VIII Standards And Design Requirements For Pressure Vessels**
- 2. Structural Design And Calculation Of Pressure Vessels**
- 3. Introduction To Pressure Vessels**
- 4. Testing And Inspection Of Pressure Vessels**
- 5. Safety, Environmental, And Occupational Health Requirements For Pressure Vessel Operation**
- 6. Introduction To PVElite Software**
- 7. Real Case Study: Design, Calculation, And Reporting Using PVElite**